

B.Tech. Degree III Semester Regular/Supplementary Examination November 2023

CS 19-202-0304 OBJECT ORIENTED PROGRAMMING (2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcomes

On successful completion of the course, the students will be able to:

- CO1: Find the basics of OOPS and relate object oriented approach for design software.
 CO2: Demonstrate the adeptness of various object oriented concepts in developing solutions to problems.
 CO3: Design and implement efficient programs for a given problem by incorporating features such as encapsulation, abstraction, inheritance etc.
 CO4: Analyse the polymorphic behaviour of objects both in run time and compile time.
 CO5: Choose between the different inheritance structures according to the problem and practice reusability.
 CO6: Experiment with generic programming and exception handling capability of C++.
 CO7: Learn the features and usage of file handling statements in C++.
- Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create
 PO – Programme Outcome

PART A

(Answer ALL questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) Differentiate between formatted and unformatted console I/O operations in C++ with suitable examples.	3	L3	1,2	1,2,3	
(b) Define and explain the concept of friend functions in C++. How do they differ from member functions, and in what scenarios are they useful?	3	L3	1,2	1,2	
(c) What are objects? How are they created?	3	L2	2	1,2	
(d) Differentiate between normal member function and const member function.	3	L2	2	1,2	
(e) What are constructors? Give an example.	3	L2	4	1,2	
(f) Explain the use of 'this' pointer?	3	L1	5	1,2	
(g) Explain how files are opened and closed.	3	L2	7	1,2,3	
(h) What is the purpose of file pointer?	3	L2	7	1,2,3	

PART B

(4 × 12 = 48)

II. (a) Discuss the features and advantages of Object Oriented Programming.	7	L2	1	1,2,3
(b) Write a C++ program to print and find the sum of first 20 prime numbers.	5	L2	2	1,2

OR

III. (a) Explain in detail about friend function.	4	L2	2	1,2
(b) Write a C++ program to use friend function acting as bridge between two classes (sum of internal marks and university marks stored of a student stored in two separate classes for 5 subjects). Prepare a mark list for the student.	8	L3	3	1,2,3

(P.T.O)

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		Marks	BL	CO	PO
IV. (a)	Explain operator overloading. Give examples each for overloading unary and binary operators.	7	L2	4	1,2
(b)	Create a class with both static and non-static members, and demonstrate the use of static member functions to perform class-wide operations.	5	L2	3	1,2,3
OR					
V. (a)	Explain the concept of abstract classes. What are its uses?	5	L2	3	1,2
(b)	Create a class in C++ having three members (an integer number, sum of digits of the number and a single digit formed by finding the sum of digits of the sum until the number is a single digit). Include appropriate functions to enter the number from user end, do calculations and display. Also include appropriate constructors for initialising all members to zero. (eg: 1236, sum of digits = 12, sum of digits as single digit = 3). Write a main program to test the class.	7	L3	3	1,2,3
VI. (a)	How can you create pointers to objects in C++? Explain how pointers can be used to access members of a class.	7	L2	4	1,2,3
(b)	When do we make a virtual function pure? Give example.	5	L3	4	1,2,3
OR					
VII.	Explain inheritance and give the different types of inheritance in C++. Also explain public, private and protected access controls.	12	L2	5	1,2,3
VIII.	Discuss in detail the different functions used to manipulate strings.	12	L2	4	1,2,3
OR					
IX. (a)	Discuss in detail about exceptional handling in C++.	6	L2	6	1,2,3
(b)	Write a C++ program to read multiple lines to a file. Copy the contents into a new file.	6	L2	7	1,2,3

Bloom's Taxonomy Levels

L1 = 4%, L2 = 70%, L3 = 26%.
